

Analysis of Correlations Among the Parameters of the Kaya Identity and the Impact of Renewable Energy on Environmental Sustainability

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Abstract—This report analyzes the influence of the parameters of the Kaya identity, their mutual correlations, and the relationship between the parameter f — carbon intensity — and an index external to the Kaya identity, namely the percentage of renewable primary energy relative to total primary energy.

I. INTRODUCTION

The Kaya identity is a mathematical identity that aims to model anthropogenic CO₂ emissions, or more generally greenhouse gas emissions. It is developed starting from the tautological identity $x = x$, where x is the amount of CO₂ emitted, which can be identified with the letter C ; consequently, $C = C$. At this point, one is free to multiply both sides by 1. Specifically, the left-hand side of the equation is left unchanged, while the right-hand side is multiplied by $\frac{P}{P}$, where P is the population. Mathematically, the identity remains unchanged, and the same reasoning can be repeated for two additional factors: GDP — Gross Domestic Product — and E — primary energy produced. Thus, one obtains:

$$C = \frac{P}{P} \times \frac{GDP}{GDP} \times \frac{E}{E} \times C \quad (1)$$

Rearranging the terms:

$$C = P \times \frac{GDP}{P} \times \frac{E}{GDP} \times \frac{C}{E} \quad (2)$$

In this way, the remaining three Kaya parameters, in addition to P , are obtained. In this report, for the sake of brevity, they are defined as:

$$g \equiv \frac{GDP}{P}; \quad e \equiv \frac{E}{GDP}; \quad f \equiv \frac{C}{E}; \quad (3)$$

g represents GDP per capita. It expresses the total value of all goods and services produced in a country over a given period of time — in this report, one year — divided by the total population of that country. This indicator is often used to measure the degree of well-being of a country's population.

e represents energy intensity, a measure that indicates the amount of energy required to produce a certain quantity of goods or services. In other words, it represents the energy efficiency of an economic activity or production process: the lower the value of e , the greater the efficiency. In the context of this report, it refers to the energy efficiency of a country.

Finally, f represents carbon intensity, a measure that indicates the amount of carbon dioxide emitted per unit of energy

produced. In this case as well, a low value denotes virtuous energy-related performance.

These four parameters are not completely independent of one another; on the contrary, they show more or less complex correlation structures. Moreover, it is not surprising that they are correlated with other indicators that are not included in the Kaya identity but are equally significant in describing the phenomenon of CO₂ emissions. In this regard, assuming that renewable energy sources have a relevant impact on emissions — a topic widely demonstrated in several articles^[1] — one may expect to identify a correlation with the Kaya parameters. Indeed, two types of correlations between f and renewable energy can be observed. First, there is a relationship between the percentage variation of f and the variation in the percentage of renewable energy relative to total primary energy over the years. Second, there is a relationship between the value of f and the percentage of renewable energy in a given country.

II. ANALYSIS AND METHODS

For the analysis of the various Kaya parameters, a data set^[2] was used containing data on the four parameters for 165 different countries, with a time range from 1965 to 2018 for some countries and from 1980 to 2018 for others. These parameters have considerably different orders of magnitude: $P \approx 10^5 - 10^8$, $g \approx 10^3 - 10^4$, $e \approx 1$, $f \approx 10^{-1}$. For this reason, percentage variations will often be used in order to highlight certain aspects.

The first aspect worth emphasizing is the influence of these parameters within the Kaya model, namely which of the four has most commonly acted as the “driving factor”. By “driving factor”, we mean the determining factor: the parameter that has increased or decreased the most relative to its initial value and relative to the time elapsed, thus significantly affecting the variation in CO₂ emissions over time. To determine this, it is useful to calculate the $CAGR$, the Compound Annual Growth Rate. The average $CAGR$ values for the 165 countries are:

$$CAGR(P) = 1.64\% \quad (4)$$

$$CAGR(g) = 2.03\% \quad (5)$$

$$CAGR(e) = -0.48\% \quad (6)$$

$$CAGR(f) = -0.35\% \quad (7)$$

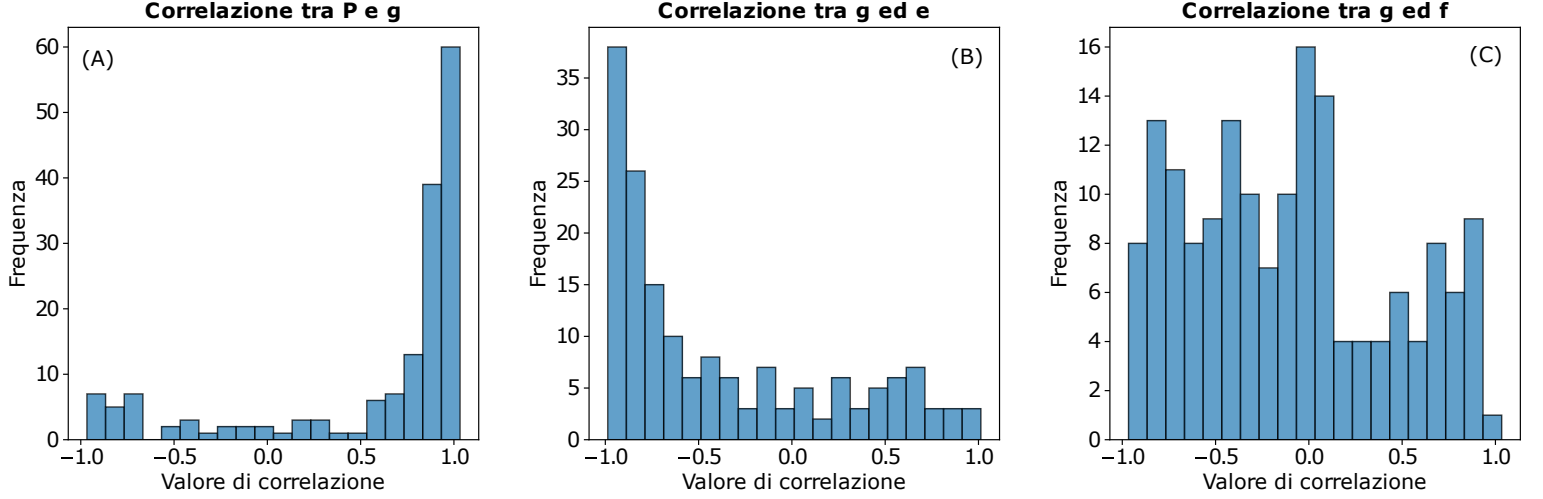


Figure 1. Correlations among the parameters P , g , and e . The correlation value between the time series of two parameters for each country increases the frequency of the interval to which it belongs by one. The strong correlation between P and g , and the strong anti-correlation between g and e , can be clearly observed.

Consequently, on average, the most influential parameter was g , which shows an annual growth of 2.03 percentage points.

It is also certainly interesting to count how many times each factor turned out to be the most influential one. The results show that in 73 countries the “driving factor” was g , in 51 countries it was P , in 35 countries it was e , and in only 6 countries it was f . Although this last category is the least represented numerically, it actually hides some suggestive insights regarding the previously mentioned relationship between renewable energy and carbon intensity f , a topic that will be discussed later.

Focusing instead on the mutual correlations among the parameters in the identity, correlation structures between P and g , and between g and e , clearly stand out. This is evident in panels (A) and (B) of Figure 1.

It should be specified that none of the countries analyzed showed a decrease in population, meaning that the population in 2018 was always greater than the population in 1965 or 1980; the same applies to g .

By calculating the correlation between the time series of parameters P and g for each individual country, it is clear that the two parameters tend to be strongly correlated with each other — panel (A), Figure 1. This implies that in the vast majority of cases, an increase in P was always accompanied by an increase in g . If the parameter g is broken down, one may conclude that an increase in P is associated with a superlinear increase in GDP relative to P itself.

The correlation between g and e is certainly more interesting. In this case as well, the correlation structure is evident and shows that, in general, the two parameters are strongly anti-correlated — panel (B), Figure 1. The increase in the parameter g has generally corresponded to a decrease in e . This means that the increase in the amount of goods and services available to each member of the population is linked to improved energy efficiency in the country where they live,

which is equivalent to a lower financial cost to produce one unit of primary energy.

As regards the correlation structure between e and f , the relationship is difficult to interpret from the graph — panel (C), Figure 1. As a first analysis, comparing the data with a Gaussian distribution gives a p -value $= 1.52 \times 10^{-5}$, which forces us to reject the hypothesis of a generally null correlation. A frequency peak around 0 can certainly be distinguished, but rather pronounced tails can also be observed.

The lack of a clear relationship between these two variables is due to their strong dependence on the energy policies adopted by each country. The effect of a change in one variable does not translate unambiguously into a change in the second variable, since the mutual “responses” to changes in a parameter are not globally encoded. A useful example is the Khazzoom-Brookes postulate^[3], an economic theory according to which an improvement in energy efficiency stimulates consumption and proportionally increases emissions.

One would therefore expect this “rebound effect” to be marked for countries that do not have renewable energy, and instead negligible for those countries that make substantial use of it. For the same increase in consumption in the two cases, a more “green” country should still emit less.

To verify this, countries with a negative percentage variation of e — comparing the last available year with the first — were considered, and the sample was divided into countries with positive correlation and countries with negative correlation with respect to f . The result is that countries with positive correlation are those with an average variation in the percentage of renewable sources of 16%, while those with negative correlation recorded a variation of 2%.

Similarly, countries with positive correlation recorded an average renewable energy percentage of 16%, while countries with negative correlation recorded 3% — both percentages calculated for the last available year.

In this sense, the correlation between e and f proves to

be complex and dependent on external factors, such as the percentage of renewable energy produced by a country.

We now focus on this aspect.

For the analysis of the relationship between carbon intensity and renewable energy, a second data set^[4] was implemented, collecting the necessary information for some countries. Other countries, previously taken into consideration, do not have data available regarding renewable energy. The combination of the two data sets led to the examination of 71 different countries.

The analysis shows that carbon intensity is strongly influenced by the percentage of renewable primary energy produced by a country. To highlight this relationship, the variations in the percentage of renewable energy of the countries that use it — indicated as $\Delta\%$ — were calculated. These variations were then organized into 5 groups: $\Delta\% < 0$, namely those countries that decreased their contribution of renewable energy; $0 < \Delta\% \leq 10$; $10 < \Delta\% \leq 20$; $20 < \Delta\% \leq 30$; and $\Delta\% > 30$. For each subgroup, the average percentage variation of f over the same time span was calculated, along with its standard deviation. It is evident that an appreciable variation in the percentage of renewable energy translates into a variation in f . The more a country increases its renewable energy portfolio, the greater the reduction in carbon intensity. The result can be seen in Figure 2.

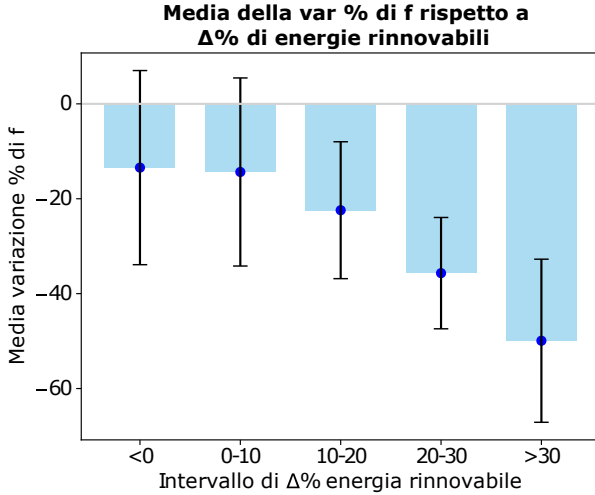


Figure 2. Trend of the percentage variation of f with respect to the variation in the percentage of renewable energy produced by a country. Each bar represents all countries falling within the $\Delta\%$ interval indicated on the x-axis. The trend of the percentage variation is clearly decreasing.

Not only has the increase in renewable energy production contributed significantly to the reduction of carbon intensity, but countries with higher percentages of renewable energy use generally record lower values of f . In other words, as the percentage of renewable energy increases, the value of f tends to decrease. This result can be observed in Figure 3, where countries with a non-zero percentage of renewables are represented. A linear fit is also proposed; the line shown has an angular coefficient $m = -0.0023$. To validate the model, the coefficient of determination was calculated, resulting in $R^2 = 0.521$, and a Student's t-test was performed, yielding a $p\text{-value} < 10^{-3}$.

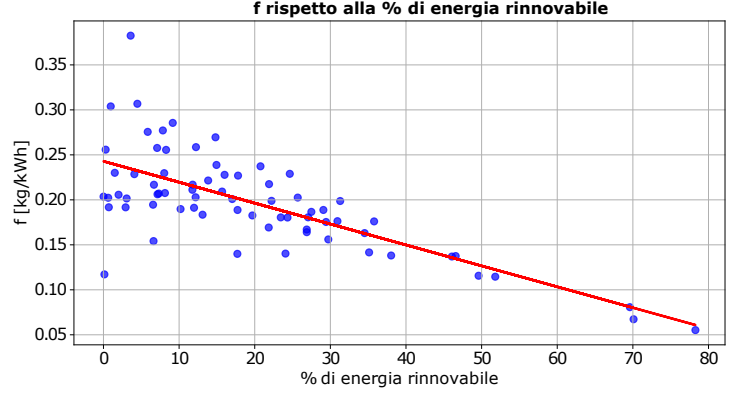


Figure 3. Scatter plot of f as a function of the percentage of renewable energy produced by the corresponding country. The red best-fit line highlights a decreasing trend of f as the percentage increases.

III. CONCLUSIONS

The Kaya identity presents some clearly evident correlation structures, such as the one between P and g or the one between g and e , while others, such as the one between g and f , are less interpretable and less clear. This is undoubtedly due to the fact that limiting the analysis to the parameters themselves, without considering external variables outside the Kaya identity, is too restrictive and constraining to provide a broader view.

For this reason, it is instructive and useful to compare the parameters with other types of variables.

The relationship between carbon intensity and renewable energy production is certainly not surprising. However, it is interesting to note how alternatives to coal and oil affect the model in two ways. The first is through variation over time, which causes f to vary accordingly. The second is in absolute terms: a high percentage of renewable energy substantially reduces f .

In both cases, the result is a significant improvement in the country's environmental sustainability.

IV. SOURCES

- [1] Environmental implications of economic complexity and its role in determining how renewable energies affect CO2 emissions
- [2] Our World In Data, data set of Kaya parameters
- [3] Harry Saunders, The Khazzoom-Brookes Postulate and Neoclassical Growth, in Energy Journal, vol. 13, no. 4, October 1, 1992, pp. 130–148.
- [4] Our World In Data, data set on renewable energy.